

Question #1 of 26

Question ID: 1586356

Which of the following *least accurately* identifies a type of publicly traded real estate security?

- A) Investment trusts.
- B) Direct mortgage lending.
- C) Operating companies.



Explanation

The main types of publicly traded real estate securities are REITs (real estate investment trusts), REOCs (real estate operating companies), and RMBS and CMBS (residential and commercial mortgage-backed securities). Direct mortgage lending is most likely to be a private rather than public investment.

(Module 33.1, LOS 33.a)

Question #2 of 26

Question ID: 1586357

Which of the following is *most likely* to represent a publicly traded real estate debt investment?

- A) A real estate operating company (REOC).
- B) Secured bank debt collateralized by real estate.
- C) A mortgage real estate investment trust (Mortgage REIT).



Explanation

Mortgage REITs are publicly traded securities that make loans secured by real estate, therefore they are publicly traded debt investments. REOCs are classified as equity (not debt) securities, while bank debt is classified as a private rather than public investment.

(Module 33.1, LOS 33.a)

Question #3 of 26

Question ID: 1586375

Patricia Ly, CFA, is a portfolio manager who wishes to add diversification to her portfolio through the addition of a real estate investment. Ly finds the following data for a particular industrial REIT:

Net operating income (NOI): \$710,000

Funds from operations (FFO): \$630,000

Assumed cap rate: 6%

Shares outstanding: 90,000 shares

Storage property average P/FFO multiple: 13x.

Industrial property average P/FFO multiple: 10x.

Ly decides to perform a valuation on this REIT. The value per share of this REIT using a price-to-FFO approach is closest to:

A) \$91



B) \$112



C) \$70



Explanation

$\text{FFO/share} = \text{FFO} / \text{Shares outstanding} = \$630,000 / 90,000 \text{ shares} = \$7/\text{share}.$

The relevant subsector average P/FFO multiple is the value for industrial properties of 10x.

$\text{FFO/share} \times \text{P/FFO multiple} = \$7.00 \times 10x = \$70.00$

(Module 33.2, LOS 33.h)

Question #4 of 26

Question ID: 1586362

Which of the following is *least likely* a difference between real estate investments and traditional asset classes like stocks and bonds?

A) Real estate tends to be indivisible



B) Real estate tends to be difficult to value



C) Real estate tends to be homogenous



Explanation

Investment in real estate is complicated by difficulty in valuing real estate, indivisibility of real estate investment (high unit value) and heterogeneity of different real estate properties even within the same class/geographical location.

(Module 33.1, LOS 33.b)

Question #5 of 26

Question ID: 1586366

Compared to transaction-based indices used to track the performance of private real estate, appraisal-based indices are *most likely* to exhibit an apparent:

- A) time lag. 
- B) higher volatility. 
- C) higher correlation with other asset classes. 

Explanation

Appraisal-based indices tend to lag transaction-based indices, as appraised values adjust only slowly to sudden shifts in the market.

Appraisal-based indices are "smoothed" by this lag, which causes appraisal-based indices to appear to have *lower* volatility and *lower* correlation with other assets than a transaction-based index would.

(Module 33.1, LOS 33.e)

Kent Clarkson, Tony Chekov and Peter Chanwit are investment consultants for a large public pension fund. They are partners in Clarkson, Chekov and Chanwit Consulting also known as 3CC. From previous meetings with the pension board, it has been established there will be an increase in exposure to real estate for the overall portfolio. Because of the defined benefit plan's significant size and their staff's expertise, the pension fund can invest and manage all forms of real estate investments. Partners of 3CC are to recommend a form of real estate investments, and recommend potential investments.

Expected Real Estate Market Conditions

Both residential and commercial real estate prices have fallen over the last five years. This trend is not expected to persist. It is a 'buyer's market' – the current supply exceeds the current demand and prices are lower than the intrinsic value. Although interest rates have fallen to historically low rates, the volume of real estate transactions remains low. Current average 20-year commercial mortgage rates are 3.75% and expected to stay relatively flat for at least 7 more years.

Loan underwriting standards have become more stringent and loan-to-value (LTV) ratios are expected to be lower than the earlier average rate of 80%.

The four forms of real estate under consideration as an investment choice for the pension fund are:

- Private: equity option is to buy commercial properties and manage them; debt option is to directly lend to commercial property investors.
- Public: equity option is to buy equity REITs; debt option is to buy mortgage REITs or CMOs.

The following information was collected by 3CC partners to aid their analysis. The returns and standard deviations of the four possible forms of real estate investments considered are listed in Exhibit 1. Correlations of real estate index with Treasury bill returns, US aggregate bond returns and US stock returns are listed in Exhibit 2.

Exhibit 1: Returns and Standard deviation (past 20 years)

	Returns	σ
Private Equity	9.5%	6.5%
Private Debt	5.5%	8.5%
Public Equity	11.5%	21.0%
Public Debt	6.2%	22.5%
Treasuries	3.5%	0.6%

Exhibit 2: Correlation of Real Estate Index With Other Asset Classes (past 20 years)

Real Estate Index Correlations	ρ
US Treasuries	0.35
US Aggregate Bonds	-0.05
US Stocks	0.25

The partners make the following statements:

Kent Clarkson: We should eliminate the private debt option from consideration. Returns for private debt are likely to be low since interest rates are likely to remain low and the amount of underwriting that is going to be required as a lender doesn't seem worth it.

Tony Chekov: I like the equity options better than the debt options based on Clarkson's private debt expectations.

Peter Chanwit: I prefer the private option over the public option since the pension fund staff can better actively manage the real estate projects and possibly outperform the index.

The partners have identified specific REIT managers who have consistently outperformed their indices for the public option. They have also contacted potential high creditworthy borrowers in case of private debt. For the private equity option, the partners are looking at different commercial properties. They have narrowed their choices to hotels and multi-family units.

Peter Chanwit is analyzing two specific buildings. Green Oaks Hotel and Blue Ridge Apartments are next to each other; have exactly the same number of units, same amenities; were built 10 years ago by the same construction company; and managed by the same property management company. They are currently owned by different entities that are also looking to provide the financing on the following basis.

Green Oaks Hotels			Blue Ridge Apartments	
Asking Price	\$25,000,000		Asking Price	\$25,000,000
Annual NOI End of Year 1	\$2,187,500		Annual NOI End of Year 1	\$2,125,000
LTV	75.0%		LTV	70.0%
Loan Interest Rate	4.00%		Loan Interest Rate	3.50%
Monthly Debt Service	\$113,621		Monthly Debt Service	\$101,493
Loan Term	20 Years		Loan Term	20 Years
Expected Sales Price in 10 Yrs	\$30,000,000.00		Expected Sales Price in 10 Yrs	\$30,000,000.00
Principal Owed at End of 10 Yrs	\$11,222,397		Principal Owed at End of 10 Yrs	\$11,144,755

The pension fund can buy one or both buildings provided they meet the minimum criteria of a debt service coverage ratio of at least 1.50X and a levered IRR of at least 17.5%.

The indices under consideration as the benchmark for private real estate equity investing are:

- Jackson Property Index (JPI) is an appraisal based index.
- Taft's Sales Index (TSI) is a repeat sales index.

- Lincoln Hedonic Index (LHI) is a hedonic index.

Concerns regarding the index choice were verbalized at a 3CC meeting:

Kent Clarkson: I'm worried about Lincoln Hedonic Index. This index may adjust for differences in property characteristics but I'm not sure it can be effective given that some properties may not sell more than once during the index's coverage period.

Tony Chekov: I don't like the Jackson Property Index. Appraisals are estimates; there haven't been many transactions lately so I question the reliability of the returns.

Peter Chanwit: I'm not sure about Taft's Sales Index. It relies on actual transactions but there are so few sales recently so how reliable are the returns?

Question #6 - 8 of 26

Question ID: 1586351

Based on projected real estate conditions and the partners' discussion given in the vignette, 3CC's top recommendation would *most likely* be:

A) public debt.



B) private equity.



C) public equity.



Explanation

The category that 3CC would most likely recommend as first choice is private equity option. Chekov prefers equity to debt option and Chanwit prefers private over public option. Clarkson wants to eliminate private debt option. Their statements are also consistent with the real estate market expectations.

(Module 33.1, LOS 33.a)

Question #7 - 8 of 26

Question ID: 1586352

If the pension fund chooses to invest in hotels over apartments, one possible reason for this is that hotels:

A) may offer higher rates of returns because of higher operational risk.



B) are commercial properties while apartments are residential properties.



C) are not affected by cost and availability of debt capital.



Explanation

All real estate values are affected by cost and availability of capital. Apartments and other multi-family units are considered commercial real estate. Hotels require more active management making them more risky ventures as more operational expertise is needed. This additional risk requires a higher rate of return.

(Module 33.1, LOS 33.a)

Question #8 - 8 of 26

Question ID: 1586353

Which statement regarding issues with indices is *least likely* correct?

- A) Chekov's statement.
- B) Chanwit's statement.
- C) Clarkson's statement.



Explanation

Clarkson's concerns about Lincoln Hedonic Index if individual properties don't sell more than once are unfounded. Hedonic Index construction does not require multiple sales of the same property.

(Module 33.1, LOS 33.a)

Question #9 of 26

Question ID: 1586358

Which of the following *most accurately* identifies one of the disadvantages of investing in real estate through publicly traded securities? Compared to other real estate investment vehicles, publicly traded securities expose investors to:

- A) unlimited liability.
- B) more-volatile returns.
- C) inferior liquidity.



Explanation

Disadvantages of investing in real estate through publicly traded securities include the volatile returns that result from pricing that is determined by the stock market. Publicly traded real estate securities offer investors the advantages of superior liquidity, and liability that is limited to the amount invested.

(Module 33.1, LOS 33.a)

Question #10 of 26

Question ID: 1586373

A key difference between Funds From Operations (FFO) and Adjusted Funds From Operations (AFFO) is that AFFO excludes:

- A) non-cash rent while FFO does not.
- B) depreciation while FFO does not.
- C) deferred tax charges while FFO does not.



Explanation

AFFO is FFO adjusted to remove straight-line rent and to provide for leasing costs and maintenance-type capital expenditures. FFO is accounting net earnings excluding deferred tax charges, depreciation, and gains or losses on sales of property and debt restructuring.

(Module 33.2, LOS 33.h)

Question #11 of 26

Question ID: 1586361

Which of the following *most accurately* identifies non-core real estate property types?

- A) Office and industrial.
- B) Hotel and hospitality.
- C) Retail and multi-family residential.



Explanation

Hospitality properties such as hotels represent relatively risky investments because these properties do not use long-term leases and their performance may be highly correlated with the business cycle. The core commercial income-producing real estate property types are retail, multi-family, office, industrial and warehouse. These "core" property types are the main properties used to create a low-risk real estate portfolio.

(Module 33.1, LOS 33.a)

Question #12 of 26

Question ID: 1586355

Which of the following *most accurately* identifies one of the characteristics of a private equity investment in income-producing real estate?

- A) Sensitivity to the credit market.



B) Homogeneity.



C) Passive management.



Explanation

Real estate values are sensitive to the cost and availability of debt capital since large amounts of borrowing are required to purchase real estate properties. Real estate is heterogeneous, as no two properties are the same. Direct ownership of real estate properties is management intensive. Other unique characteristics possessed by real estate properties include: fixed location, high unit value, depreciation, high transaction cost, illiquidity, and difficult to value.

(Module 33.1, LOS 33.a)

Question #13 of 26

Question ID: 1586359

Which of the following *most accurately* identifies one of the advantages of investing in real estate through publicly traded securities?

A) Publicly traded corporate structures cost less to maintain.



B) Diversification by geography and property type is facilitated.



C) Structural conflicts of interest are eliminated.



Explanation

One of the advantages of publicly traded real estate securities is that they offer investors greater potential for diversification by geography, property, and property type. Disadvantages of publicly traded real estate securities include the costs of maintaining a publicly traded corporate structure, and the potential for structural conflicts of interest that can occur between the partnership and REIT shareholders under an UPREIT or DOWNREIT structure.

(Module 33.1, LOS 33.a)

Question #14 of 26

Question ID: 1586367

A real estate market is characterized by frequent transactions. However, individual properties have long holding periods. Which real estate pricing index would be *least* suitable in such an environment?

A) Repeat sales index.



B) Hedonic price index.



C) Appraisal based index.



Explanation

Repeat sales index relies on repeat sales of individual properties. Since individual properties have long holding periods, repeat sales index would be least suitable. Hedonic price index relies on transaction data and the regression model explains the variation in transaction prices based on differences between individual properties sold. Appraisal based indices use transaction prices also to estimate value after adjustments for differences. Since there are plenty of transactions, appraisal and hedonic price index have sufficient data to provide good value estimates.

(Module 33.1, LOS 33.e)

Question #15 of 26

Question ID: 1586363

Retail sales growth is *most likely* to be a top economic factor affecting the economic value of a(n):

A) industrial REIT.



B) residential REIT.



C) health care REIT.



Explanation

Other than retail REITs, retail sales growth is an important factor driving economic value of industrial REITs.

(Module 33.1, LOS 33.b)

Question #16 of 26

Question ID: 1586372

The net asset value approach to valuation makes sense for REITs because:

A) NAV equals the value that public equity investors attach to a REIT.



B) there exist active private markets for real estate assets.



C) the price at which a REIT trades very closely tracks NAV.



Explanation

Because active private markets for real estate assets exist, REITs lend themselves to a net asset value approach to valuation. NAV reflects the estimated value of REIT assets to a private market buyer, however this may be different from the value that public equity investors would attach to the REIT. REITs have historically traded at a large premium or discount to NAV.

(Module 33.2, LOS 33.g)

Question #17 of 26

Question ID: 1586376

Which of the following *most accurately* describes an approach to REIT valuation?

- A) The discounted cash flow approach typically consists of intermediate-term cash flow projections plus a terminal value based on cash flow multiples. 
- B) The P/AFFO approach avoids estimates and assumptions in its calculation. 
- C) The P/FFO approach adjusts for the impact of recurring capital expenditures needed to keep properties operating smoothly. 

Explanation

In discounted cash flow REIT models, investors generally use intermediate-term cash flow projections and a terminal value based on historical cash flow multiples. FFO does not adjust for the impact of recurring capital expenditures needed to keep properties operating. AFFO adjusts for routine maintenance type capital expenditures, but assumptions and estimates (which may vary widely) are required in the calculation of AFFO.

(Module 33.2, LOS 33.i)

Question #18 of 26

Question ID: 1586360

Which of the following is the *most likely* to represent an advantage of investing in publicly traded real estate securities over direct ownership of property? Publicly traded real estate securities offer:

- A) more control over investment decisions. 
- B) greater liquidity. 
- C) lower price volatility. 

Explanation

One of the main advantages of investing in publicly traded equity real estate securities stems from the fact that these securities trade on stock exchanges, which results in greater liquidity compared with buying and selling real estate directly. The downside of trading on a stock exchange is that publicly traded equity real estate securities have greater price volatility than do directly owned properties. Another disadvantages of publicly traded real estate securities is that they offer investors little to no control over investment decisions.

(Module 33.1, LOS 33.a)

Question #19 of 26

Question ID: 1586374

Which of the following is an expense normally deducted from accounting net earnings but not from FFO?

A) Property operating expenses



B) Depreciation expense



C) Property taxes



Explanation

Depreciation on real estate is excluded from FFO because most investors believe that real estate maintains its value to a greater extent than does other types of long-term business assets. Therefore, taking depreciation deductions, which reduce the value of the real estate, does not represent economic reality. FFO is accounting net earnings excluding depreciation charges on real estate, deferred tax charges, and gains or losses from sales of property and debt restructuring. Property operating expenses and property taxes are both normal rental expenses deducted to arrive at operating income.

(Module 33.2, LOS 33.h)

Question #20 of 26

Question ID: 1586371

If a REIT has assets with a current market value of \$3,000,000, liabilities with a current market value of \$2,000,000, and 100,000 shares outstanding, what is the NAVPS per share?

A) \$50.00



B) \$30.00



C) \$10.00



Explanation

NAVPS per share can be calculated by beginning with assets, subtracting liabilities, and then dividing the result by the number of shares outstanding. Thus,
 $\$3,000,000 - \$2,000,000 = \$1,000,000$ and $\$1,000,000 / 100,000 = \10.00 per share.

(Module 33.2, LOS 33.g)

Question #21 of 26

Question ID: 1586364

Appropriate due diligence in a private real estate investment is *most likely* to:

A) mitigate unforeseen potential problems.



B) lower existing operating costs.



C) review lease and rental history.



Explanation

Due diligence can be very costly but it can potentially lower risk of unexpected legal and physical real estate problems. Due diligence will usually increase current operating costs. A review of lease and rental history is one example of due diligence not a possible result of due diligence.

(Module 33.1, LOS 33.d)

Question #22 of 26

Question ID: 1586368

Compared with REITs, real estate operating companies (REOCs) are *most likely* to feature higher:

A) yields.



B) levels of income tax exemption.



C) operating flexibility.



Explanation

REOCs have greater operating flexibility to invest in a wide range of real estate than do REITs. REITs offer higher yields compared to REOCs. REITs offer income tax exemption while REOCs generally do not.

(Module 33.2, LOS 33.f)

Question #23 of 26

Question ID: 1586354

Which of the following *most accurately* identifies a private equity investment in income-producing real estate?

- A) Investment in a real estate investment trust (REIT). 
- B) Direct ownership of real estate properties. 
- C) Private market mortgage lending by an insurance company. 

Explanation

Real estate investments take four major forms: private equity, publicly traded equity, private debt, and publicly traded debt. Private equity investment in real estate refers to direct ownership of real estate properties. Mortgage lending by banks or insurance companies is best described as private debt. Indirect ownership of real estate through equity securities such as REITs is an example of publicly traded equity.

(Module 33.1, LOS 33.a)

Question #24 of 26

Question ID: 1586369

The *most likely* consequence of the high income distribution that REITs are required to make is:

- A) dividend yields that are nearly on-par with the yields of other publicly traded equities. 
- B) high volatility of reported income. 
- C) frequent secondary equity offerings compared to other kinds of companies. 

Explanation

Because REITs are not able to retain earnings as other companies do, REITs make frequent secondary equity offerings, in order to finance growth and property acquisitions. REITs' required distributions result in a dividend yield that is significantly higher than those of most other publicly-traded equities. REITs' focus on income from rental properties leads to low volatility of reported income.

(Module 33.2, LOS 33.f)

Question #25 of 26

Question ID: 1586370

When calculating NAVPS, a real estate company's assets and liabilities are valued at their:

A) book value.



B) liquidation value.



C) market value.



Explanation

All assets and liabilities of a company are taken at current market value when calculating NAVPS. NAVPS is a superior measure of a company's net worth when compared to its book value per share.

(Module 33.2, LOS 33.g)

Question #26 of 26

Question ID: 1586365

Which of the following *least accurately* describes a major category of due diligence factors that should be investigated in determining the value of a property?

A) Operating expenses.



B) Structural integrity.



C) Pipeline analysis.



Explanation

The major due diligence factors that are likely to affect the value of a property include: operating expenses; structural integrity; environmental issues; leases and lease history; lien, ownership, and property tax history; and compliance with relevant regulations and laws.

(Module 33.1, LOS 33.d)